

One Line, Two Model Diagnostics

AP Statistics · Unit 5 · Topic 5.4 · TEACHER KEY

CED TETHER

- ENDURING UNDERSTANDING: DAT-1 Regression models may allow us to predict responses to changes in an explanatory variable.
- SKILLS:
- **Skill 2.B** — Construct numerical or graphical representations of distributions.
- **Skill 2.A** — Describe data presented numerically or graphically.
- LO DAT-1.E Represent differences between measured and predicted responses using residual plots. [Skill 2.B]
- EK DAT-1.E.1 The residual is the difference between the actual value and the predicted value: $\text{residual} = y - \hat{y}$.
- EK DAT-1.E.2 A residual plot is a plot of residuals versus explanatory variable values or predicted response values.
- LO DAT-1.F Describe the form of association of bivariate data using residual plots. [Skill 2.A]
- EK DAT-1.F.1 Apparent randomness in a residual plot for a linear model is evidence of a linear form to the association between the variables.
- EK DAT-1.F.2 Residual plots can be used to investigate the appropriateness of a selected model.

Essential question: How should correlation and residuals be reconciled when judging a model for safety planning?

DOK-3 skill: 4.B · **FRQ pattern:** correlation-residuals-model-choice-and-scope

DOK rationale: Part (c) weighs competing claims using the day's numerical or graphical evidence and explains a limitation or consequence; the judgment requires linked reasoning beyond a calculation.

Self-paced bonus sheet. Students take it when they choose and turn it in when they finish. Everything they need is printed on the sheet. Score part (c) only, E / P / I, by hand — never AI-graded or auto-scored. (a) and (b) are the ladder up; use them to see where a thin (c) came from.

Questions to ask

- Which number or visible feature supports your claim?
- Why does the other claim go beyond that evidence?
- What would you tell someone who chose the other claim?

[WARN] LOOK FOR

- A choice with copied numbers but no explanation of how they support it.
- Treating a model or average as a guaranteed individual outcome.

SCORING PART (c) — E / P / I

Expected elements

- **reasoning-1** (required) — Supports June by reconciling the large correlation with the U-shaped residuals
- **reasoning-2** (required) — Links the 4-foot underprediction to a stopping-distance consequence

- **reasoning-3** (required) — Requires a replacement residual check in the stated vehicle/surface setting

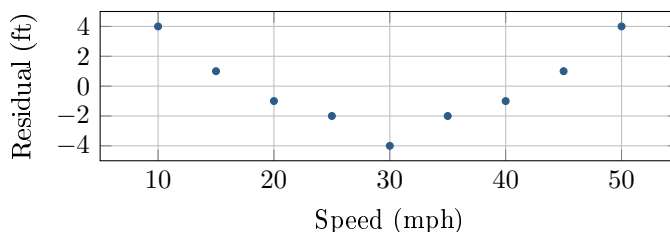
E	Includes all three required reasoning elements above, with correct contextual evidence.
P	Includes at least one substantive correct reasoning link above, but omits or misstates another required element.
I	Includes no substantive correct reasoning link above; an unsupported choice or copied number alone is insufficient.

[STAR] THE PROBLEM — annotated

Suppose a road-safety lab records braking distance y at speeds x from 10 through 50 mph on one vehicle and dry surface. The line is $\hat{y} = -10 + 2.5x$, the correlation is $r = 0.997$, and the residual plot is shown. A residual is actual distance minus the line's prediction. At 50 mph, the actual distance is 119 feet.

Ezra: The large correlation and residuals on both sides of 0 justify using the line across the tested speeds.

June: I would use where the residuals occur and the 50-mph error to judge safety, then check any replacement model separately.



FIRST TAKE — one sentence before you work the parts

Would you trust this line to plan a safe stopping distance? Give one number or visible pattern in one sentence.

Key: Not graded. Students should decide what each diagnostic establishes before judging the line.

[BOOK] READING A RESIDUAL PLOT (keep on the page)

A residual is $y - \hat{y}$. A positive residual means the model underpredicted; a negative residual means it overpredicted. Plot residuals against x or $\hat{y}$. Random scatter around 0 supports a linear form. Curvature, trends, clusters, or changing spread show structure the line missed. Correlation measures the direction and strength of linear association, but even a large $|r|$ does not replace checking residual locations. Check a replacement model with its own residual plot.

DOK 1 (a) Read the residual at 50 mph. Does the line predict too high or too low there?

Key: The residual is 4 feet, so the line predicts too low.

DOK 2 (b) Verify the 50-mph residual using the equation. Describe where the residual plot is above and below zero.

Key: At 50 mph, $\hat{y} = -10 + 2.5(50) = 115$ feet, so the residual is $119 - 115 = 4$ feet. Residuals are positive at 10–15 and 45–50 mph, and negative at 20–40 mph, forming a U-shaped pattern.

DOK 3 (c) Whose judgment is supported? Explain why the residual pattern matters even with the large correlation, and connect the 50-mph error to safety. State what you would check in a replacement model for this vehicle and dry surface. Write 3–4 sentences.

Key: June is supported: $r = 0.997$ shows a strong positive linear association, but the U-shaped residuals show a pattern of errors the line misses. Having both signs is not enough when the signs occur in a predictable order. At 50 mph the 4-foot underprediction could make a stopping zone too short. For this vehicle and dry surface, check that a replacement model's residuals have no clear pattern around zero before trusting it.